

NovaCart Grounded Metrics Agent

A retrieval-augmented n8n workflow that answers leadership metrics questions strictly from internal documents — or refuses, rather than guesses.

n8n

Pinecone

OpenAI GPT-4.1

Cohere Rerank

Google Drive

The Problem

NovaCart leadership needed trustworthy answers from data that was scattered, slow to compile, and easy for a generic LLM to fabricate.

1

Data Silos

Weekly metrics — revenue, conversion, churn, SLA — sit locked inside separate Word and PDF documents in Drive.

2

Slow Turnaround

Analysts manually open files, hunt for numbers, and hand-write summaries before every leadership review.

3

Hallucination Risk

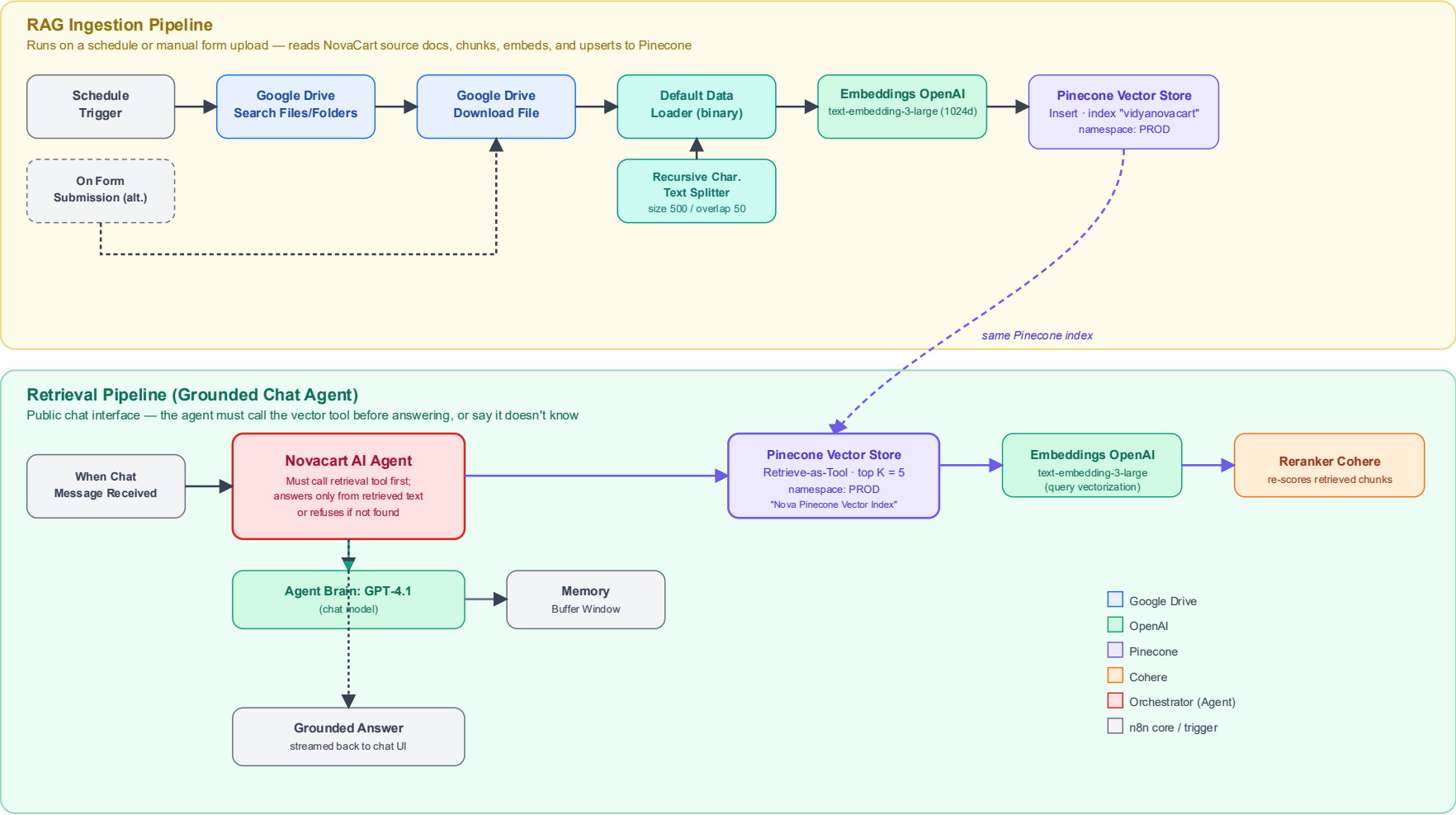
Generic ChatGPT-style tools invent plausible numbers when they don't actually have NovaCart's private data.

System Flow

Two linked n8n pipelines share one Pinecone index: ingestion keeps it fresh, retrieval answers only from it.

NovaCart RAG Ingestion & Retrieval Pipeline (n8n)

Two connected n8n workflows: document ingestion into Pinecone, and a grounded chat agent that retrieves from it



Key Decisions & Trade-offs

1

Force one retrieval call before any answer

The agent's system prompt requires it to call the Pinecone tool exactly once per question, before it's allowed to respond.

2

Exact refusal string on empty retrieval

If the index has no relevant chunk, the agent must return one fixed refusal line — not an improvised apology or guess.

3

Trade-off: auditability over flexibility

This makes some edge-case questions less conversational, but a predictable “I don’t have that” beats a confident wrong number in a leadership tool.

4

Cohere reranking before the model sees chunks

Top-5 vector matches are re-scored by Cohere so the agent reads higher-precision context, not just raw cosine-similarity order.

Outcome

Documents available of a real, executed run (see docs/Screenshots.pdf).

What Held Up

- Full end-to-end execution confirmed: ingestion populated Pinecone, and three separate chat prompts returned grounded, correct answers.
- Forcing one tool call + an exact refusal string made “I don’t know” predictable and easy to test.
- Reranking measurably tightened up which chunks the agent actually saw.

What's Next

- Widen ingestion beyond Google Drive to Notion/Confluence, as named in the original problem statement.
- Add authentication in front of the public chat webhook before any real internal rollout.
- Surface citation metadata directly in the agent's final answer, not just in its retrieval context.